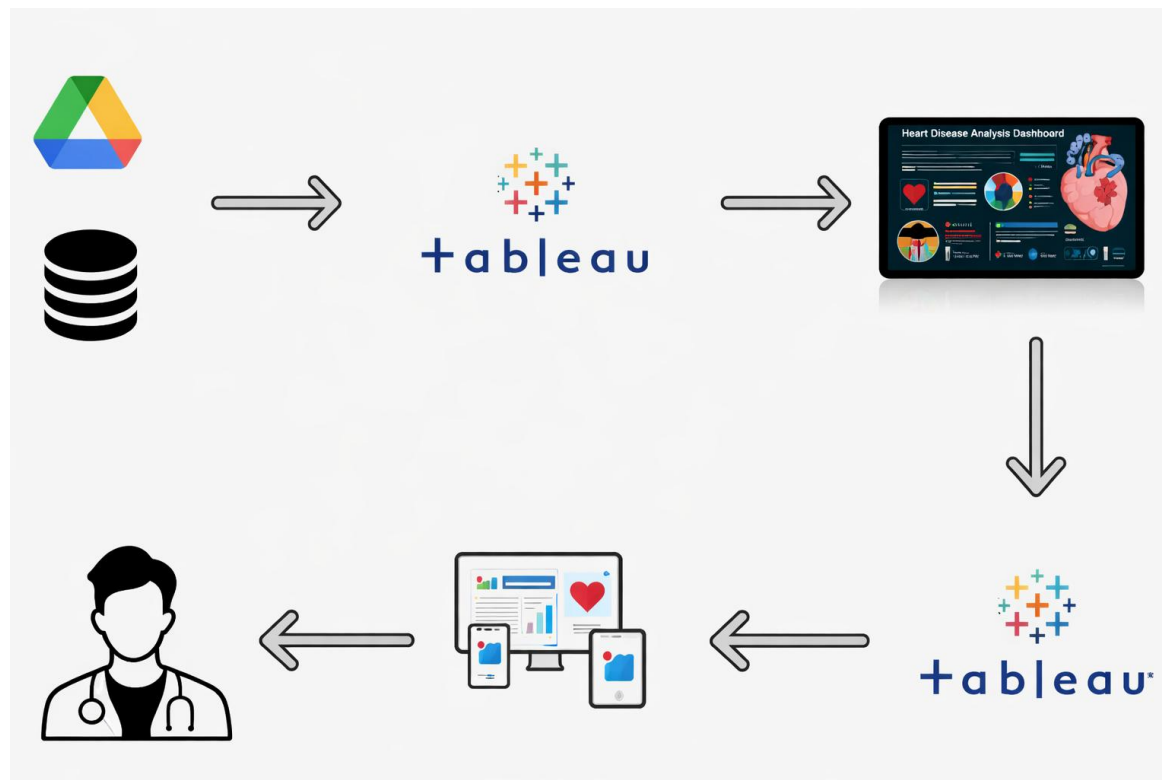


Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	16 February 2026
Team ID	LTVIP2026TMIDS63806
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Technical Architecture:



Technology Stack

Below is the technology stack used in the Heart Disease Analysis Project

Table-1: System Components

Sno	Components	Description	Technology
1	User Interface	Interface through which healthcare professionals interact with the dashboard (Web UI / Mobile View).	HTML, CSS, JavaScript / React JS / Angular JS
2	Data Integration Layer	Connects and imports heart disease datasets from CSV files, hospital databases, or EMR exports.	Python, Tableau
3	Data Preparation & Transformation	Cleans, formats, and prepares raw patient data (handling null values, correcting data types, categorizing risk levels).	Tableau Prep, Python (Pandas, NumPy)
4	Analytics & Visualization Engine	Creates charts, KPIs, and dashboards to visualize patient demographics and risk factors.	Tableau Desktop
5	Calculated Metrics & Clinical Logic	Computes KPIs such as Risk Percentage, Average Cholesterol, BP Categories, and High-Risk Classification.	Tableau Calculated Fields, Python
6	Database	Stores structured heart disease datasets and processed data extracts.	MySQL / PostgreSQL / CSV Data Sources
7	Cloud & File Storage	Stores datasets and dashboard assets for sharing and backup.	Google Drive / AWS S3 / Local Storage
8	Dashboard Publishing Platform	Publishes dashboards for hospital staff or public demonstration.	Tableau Public / Tableau Server
9	External Data Sources	Integrates additional datasets such as lifestyle data, WHO statistics, or public health datasets.	Open Government Data APIs / CSV Datasets
10	Performance Optimization	Improves dashboard responsiveness and data refresh performance.	Tableau Extracts (.hyper), Data Indexing
11	Infrastructure & Deployment	Hosts dashboards for access by healthcare stakeholders and ensures availability.	Local Server / Cloud Hosting

Table-2: Application Characteristics:

S.No	Category	Description	Technology Used
1	Open-Source Frameworks	Libraries used for data cleaning, analysis, and predictive modeling of heart disease risk.	Python, Pandas, NumPy, Scikit-Learn
2	Security Implementations	Protects sensitive patient data and controls dashboard access.	Role-Based Access Control (RBAC), OAuth 2.0
3	Scalable Architecture	Supports addition of new patient records, risk factors, and historical health datasets.	Cloud Storage, REST APIs
4	Availability	Ensures dashboard is accessible during clinical reviews and stakeholder meetings.	Tableau Server Deployment / Cloud Hosting
5	Performance	Optimized data extracts and filtering for fast dashboard loading and interaction.	Tableau Extracts (.hyper), Indexed Queries
6	Data Accuracy	Validates KPIs and risk calculations against raw patient datasets before publishing.	Data Validation Scripts (Python)
7	Usability	Clean layout with consistent color coding for risk levels (High/Medium/Low) and readable medical metrics.	Tableau Design Principles
8	Reliability	Handles missing or incomplete patient data without affecting risk calculations or dashboard performance.	Error Handling in Python / Data Cleaning Logic